JAMES CHRISTOPHER HARTLEY

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PROFESSIONAL SUMMARY

Data scientist with 12 years of experience building machine learning systems and predictive models for large-scale financial and retail organisations. Started career in statistical analysis and progressed through multiple senior technical roles, leading data science teams and architecting end-to-end ML pipelines. Passionate about solving complex business problems through applied mathematics and writing clean, maintainable code. Currently based in Birmingham and actively seeking challenging roles in advanced analytics and AI development.

WORK EXPERIENCE

Senior Data Scientist

DataVault Solutions

https://www.datavaultsolutions.io

London, United Kingdom

March 2020 - Present

Architected and deployed 12 machine learning models in production that improved fraud detection accuracy by 34%, saving the company approximately 2.1 million pounds annually in prevented losses.

Led a team of 5 junior data scientists and 2 machine learning engineers, conducting weekly code reviews, mentoring on statistical methods, and establishing best practices for model validation and reproducibility.

Designed and implemented an automated retraining pipeline using Apache Airflow that reduced model decay issues by 67% and decreased manual intervention from 40 hours per month to just 8 hours.

Conducted end-to-end analysis on customer churn patterns across 15 million active users, delivering actionable insights that informed product strategy and increased customer lifetime value by 18 percent.

Lead Data Scientist

Precision Analytics Group

https://www.precisionanalytics.co.uk

Birmingham, United Kingdom

June 2016 - February 2020

Developed a gradient boosting ensemble model for demand forecasting in retail supply chain, achieving 22% improvement in forecast accuracy and reducing inventory holding costs by 5.3 million pounds over two years.

Established the company's first formal machine learning operations framework, including model governance, monitoring dashboards, and A/B testing infrastructure that became the gold standard across all analytical teams.

Built and maintained a data science knowledge base documenting 47 completed projects, methodologies and lessons learned, which was adopted by 3 partner organisations and presented at 2 industry conferences.

Managed cross-functional projects with stakeholders in finance, operations and marketing, translating technical recommendations into business strategy and securing executive buy-in for 4 major analytical initiatives worth 3.2 million pounds in investment.

Data Scientist

Quantum Insights Ltd

https://www.quantuminsights.net

Manchester, United Kingdom

September 2011 - May 2016

Developed time series forecasting models using ARIMA and LSTM neural networks to predict customer behaviour, achieving mean absolute percentage error of 11.3% across 8 different product categories.

Conducted exploratory data analysis on datasets exceeding 500 million records, identifying data quality issues and creating automated validation rules that caught 94% of anomalies before downstream processing.

Implemented natural language processing pipelines to extract insights from 2 million customer reviews, reducing manual sentiment analysis work by 85% and identifying top 20 product improvement opportunities.

Designed and delivered training workshops on statistical modelling and Python for analytics to 120 employees across the organisation, resulting in increased adoption of data-driven decision making.

EDUCATION

University of Birmingham

MSc Data Science and Machine Learning

2010 - 2011

University of Warwick

BSc Mathematics with Statistics

2007 - 2010

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Scala, Java

Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, CatBoost

Statistical Methods: Regression Analysis, Time Series Forecasting, Bayesian Inference, Experimental Design

Data Processing: Pandas, NumPy, Spark, Hadoop, Databricks

Databases: PostgreSQL, MySQL, Snowflake, MongoDB, Cassandra

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform (BigQuery, Vertex AI), Azure ML

Big Data Tools: Apache Spark, Airflow, Kafka, Hive

Visualisation: Tableau, Matplotlib, Seaborn, Plotly, ggplot2

Version Control: Git, GitHub, GitLab

MLOps: Model Registry, Pipeline Orchestration, Model Monitoring, Docker, Kubernetes

CERTIFICATIONS

AWS Certified Machine Learning Specialist - 2019

Microsoft Azure Data Scientist Associate - 2021